ASSIGNMENT #04

Course: CHE 341: CHEMICAL PROCESS CONTROL

Term: Spring 2026

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Due Date: Friday, March 20rd TA's Mailbox by 3:00PM

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a typed report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report.

Assignment #4:

1 Introduction:

In this assignment, the dynamic behavior of dissolved oxygen concentration in a bioreactor is analyzed and controlled through manipulation of air flow rate. Oxygen is transferred from gas bubbles created by a sparger into the liquid phase and simultaneously consumed by biological reactions. The system is inherently dynamic and must be characterized for control design.

Overall, the objective of this assignment is to develop a dynamic model for the time domain and the Laplace domain. The Laplace domain must be linearized and Skogested approximations must be applied. The process is simulated in which the process gain, time constant, and time delay are extracted.

The reactor consists of a gas sparging, a mixer, oxygen transfer, and biological consumption. Key mechanisms include the mass transfer term and the biological consumption term.

1. Mass transfer

$$Transfer\ rate = \ k_L a\ (C^* - C)$$

2. Biological consumption

$$R_{O_2} = K_{O_2}\ x$$

2 Dynamic Model Development

2.1 General Mass Balance:

The dynamic behavior of dissolved oxygen in the bioreactor is governed by a species mass balance on oxygen in the liquid phase. The overall general mass balance is an equation which integrates the input, output, generation, and consumption.

$$Accumulation = Rate\ of\ Input - Rate\ of\ Output + Generation - Consumption$$

For this system, there is no inlet or outlet of liquid-phase oxygen, and oxygen only enters via mass transfer from gas bubbles. Oxygen is removed via biological consumption. Therefore, the mass balance can be simplified.

$$Accumulation = Transfer\ in - Consumption$$

The accumulation of dissolved oxygen in the reactor is the volume of the reactor multiplied by the change of dissolved oxygen over time.

$$V\ \frac{dC}{dt}$$

The mass transfer term follows Newton's law of mass transfer, in which the transfer rate is the product of the driving force and volumetric mass transfer coefficient, as discussed earlier. This leads to the overall inlet transfer being the product of the mass transfer term multiplied by the volume.

Combining all of the terms leads to the derivation of the final dynamic model for the time domain.

$$\frac{dC}{dt} = k_L a \,(C^* - C) - K_{O_2}\, x$$

Using the parameter values that were given in the problem statement, the final ODE can be calculated.

$$\frac{dC}{dt} = [0.25 + 0.001(F - 500)](C^* - C) - 0.000025$$

2.2 Steady State Analysis

At steady state, the change in concentration over time is equal to zero.

$$\frac{dC}{dt} = 0$$

$$0 = k_L a \,(C^* - C) - K_{O_2}\, x$$

We can also calculate the C^* by using Henry's Law, in which Henry's constant is corelated with the partial pressure of oxygen within the stream.

$$C^* = HP_{O_2}$$

We can calculate the partial pressure of oxygen in air using the air composition given in the problem by the total pressure, which is 1. From this, a value for C^* can be calculated.

$$C^* = 0.00022\ kmol/m3$$

From this, we can see that at steady state, the oxygen entering the system is equal to the oxygen consumed. The system operated below saturation due to its biological demand, as the driving force represents equilibrium imbalance required to sustain metabolism.

3 Linearization

To obtain a transfer function, the nonlinear model must be linearized around steady state. Deviation variables are defined.

$$Y = C - C_{SS}\, , X = F_{air} - 500$$

The deviation variables were defined according to Chapter 5 methodology, where the output variable (dissolved oxygen concentration) and input variable (air flow rate) are expressed as deviations from their steady-state values to facilitate linearization and transfer function derivation.

We can substitute it into the model.

$$\frac{dY}{dt} = [0.25 + 0.001X](C^* - C_{SS} - Y) - \ 0.000025$$

Using steady state conditions and further simplification.

$$\frac{dY}{dt} + 0.25Y = 0.00000011\,X$$

With this ODE, we can determine that the system behaves as a first-order model, where the input is the airflow deviation and the output is the dissolved oxygen deviation.

4 Laplace Transform

The Laplace Transform can be taken from the linearized ODE.

$$s\,Y(s) + 0.25Y(s) = 0.00000011\,X(s)$$

The ratio between Y and X therefore becomes:

$$\frac{Y\,(s)}{X\,(s)} = \frac{0.00000011}{s + 0.25}$$

This is the final Laplace transfer function of process. It can be interpreted in order to get values for τ and Kp.

Standard form:

$$G_p(s) = \frac{K_p}{\tau s + 1}$$

Matching terms:

$$\tau = \frac{1}{0.25} = 4\text{ s}$$

$$K_p = \frac{1.1 \times 10^{-7}}{0.25} = 4.4 \times 10^{-7}$$

We also need to apply sensor dynamics to the process model function.

From assignment:

$$\tau_m = 30\text{ s}$$

Sensor model:

$$G_m(s) = \frac{1}{30s + 1}$$

Total system:

$$G(s) = \frac{4.4 \times 10^{-7}}{(4s + 1)(30s + 1)}$$

5 Skogestad Approximation.

In order to simplify the process to a first order FOPTD, we use the Skogestad approximation.

To simplify to FOPTD:

$$G(s) = \frac{K_p e^{-\theta s}}{\tau s + 1}$$

Using Skogestad:

$$\tau = 30 + \frac{4}{2} = 32$$
$$\theta = \frac{4}{2} = 2$$

Therefore, the final model becomes:

$$G(s) = \frac{4.4 \times 10^{-7} e^{-2s}}{32s + 1}$$

6 Excel Simulation

To validate the analytical model derived from the Laplace transform, a dynamic simulation was performed in Excel using the time-domain ODE for dissolved oxygen concentration. The purpose of this simulation was to observe the transient response of the process to a 10% step increase in air flow rate, then extract the process gain, lag time, and time delay directly from the simulated response and compare them with the values predicted by the linearized FOPTD transfer function. This directly addresses Part 3 of the assignment.

The simulation was based on the oxygen mass balance developed earlier:

$$\frac{dC}{dt} = k_L a(C^* - C) - K_{O_2} x$$

with

$$k_L a = 0.25 + 0.001(F_{air} - 500)$$

and

$$K_{O_2} x = (1.1 \times 10^{-4})(0.25) = 2.75 \times 10^{-5}$$

The nominal operating conditions were taken from the assignment:

$$F_{air,ss} = \frac{500 \text{ ft3}}{\text{min}}$$

$$C(0) = C_{ss} = 1.1 \times 10^{-4} \text{ kmol/m3}$$

$$C* = 2.2 \times 10^{-4} \text{ kmol/m3}$$

Both the time-domain model and the Laplace-domain model describe the same steady-state behavior, although they approach it in slightly different ways. From the time-domain plot, the dissolved oxygen concentration begins at approximately 1.1×10^{-4}kmol/m3 and increases gradually before leveling off at a steady-state value near 1.5×10^{-4} kmol/m3.

This indicates that the system was initially not at equilibrium and evolves dynamically until the condition $\frac{dC}{dt} = 0$is satisfied. In contrast, the Laplace-domain response begins at zero in deviation form and rises rapidly to a steady-state deviation of approximately 2.2×10^{-5} kmol/m3, which corresponds exactly to the difference between the final and initial values observed in the time-domain simulation. When this deviation is added back to the steady-state baseline, it yields the same final concentration as the time-domain model.

The agreement between the two models is evident in their final steady-state values, as both ultimately predict the same increase in dissolved oxygen concentration. However, the transient behavior differs noticeably. The Laplace-domain response rises more quickly and reaches its steady-state value faster, while the time-domain response exhibits a slightly slower approach. This difference arises because the Laplace-domain model is based on a linearized approximation of the system near steady state and does not fully capture the nonlinear dependence of the mass transfer coefficient $k_L a$on the air flow rate. In the original time-domain model, $k_L a$varies continuously with the operating conditions, which introduces slight nonlinear effects and results in a more gradual response.

Another important distinction is that the Laplace-domain model is expressed entirely in deviation variables. As a result, it inherently assumes that the system is initially at steady state, and any response represents a change from that equilibrium condition. This is why the Laplace-domain plot starts at zero. The time-domain model, on the other hand, tracks the absolute concentration directly and therefore shows the full evolution of the system from its initial condition to its steady state.

Despite these differences in transient behavior, both models are consistent in their physical interpretation. In both cases, the system approaches a steady state where oxygen transfer into the liquid phase balances oxygen consumption by the cells. The time-domain model demonstrates this through the gradual reduction of the accumulation term to zero, while the Laplace-domain model demonstrates it through the stabilization of the deviation variable. The fact that both models converge to the same steady-state value confirms that the linearization and Laplace transform procedures were applied correctly and that the derived transfer function accurately represents the system behavior near the operating point.

This comparison is important because it highlights the strengths and limitations of each modeling approach. The time-domain model provides a more complete representation of the system dynamics, including nonlinear effects, while the Laplace-domain model provides a simplified and analytically convenient representation that is particularly useful for control system design. The close agreement in steady-state predictions validates the use of the transfer function, while the differences in transient response emphasize the approximations involved in linearization.

The airflow was held at its nominal value until the system reached steady state. Based on the simulation, steady-state behavior was established by approximately 55 s. At that time, the airflow was increased by 10%, giving

$$\Delta F_{air} = 0.10(500) = 500 \text{ ft3/min}$$

so that the new airflow became

$$F_{air} = 550 \text{ ft3/min}$$

In Excel, the ODE was integrated numerically over time, and the dissolved oxygen concentration was plotted against time. The resulting response showed a smooth monotonic rise from the initial value to a new steady value, which is characteristic of a first-order process.

The first quantity obtained from the simulation was the process gain. From the time-domain response, the dissolved oxygen concentration increased from approximately

$$C_i = 1.10 \times 10^{-4} \text{kmol/m3}$$

to approximately

$$C_f = 1.50 \times 10^{-4} \text{kmol/m3}$$

Therefore, the total output change was

$$\Delta C = C_f - C_i$$
$$\Delta C = (1.50 - 1.10) \times 10^{-4} = 4.0 \times 10^{-5} \text{kmol/m3}$$

The corresponding process gain from the dynamic simulation is

$$K_{p,\text{sim}} = \frac{\Delta C}{\Delta F_{air}}$$
$$K_{p,\text{sim}} = \frac{4.0 \times 10^{-5}}{50} = 8.0 \times 10^{-7} \frac{kmol/m3}{ft3/min}$$

The lag time, or time constant, was estimated from the response curve using the standard first-order definition. For a first-order system, the time constant is the time required for the response to reach 63.2% of its total final change after the step is applied. Thus,

$$0.632(\Delta C) = 0.632(4.0 \times 10^{-5}) = 2.53 \times 10^{-5} \text{ kmol/m3}$$

The concentration corresponding to one time constant after the step is therefore

$$C(\tau) = C_i + 0.632(\Delta C)$$
$$C(\tau) = 1.10 \times 10^{-4} + 2.53 \times 10^{-5} = 1.353 \times 10^{-4} \text{ kmol/m3}$$

From the Excel plot, this value was reached roughly 5 s after the process began responding, so the simulated lag time was taken as

$$\tau_{\text{sim}} \approx 5 \text{ s}$$

The time delay was determined by observing whether there was any period after the step change during which the output remained flat before starting to respond. From the plotted data, the dissolved oxygen concentration began changing essentially immediately after the airflow perturbation, so the simulated delay was approximated as

$$\theta_{\text{sim}} \approx 0 \text{ s}$$

From the linearized model and Laplace transform derivation, the process transfer function without the sensor was found to be

$$\frac{Y(s)}{X(s)} = \frac{4.4 \times 10^{-7}}{4s + 1}$$

Including the sensor lag of 30 s given in the assignment produced

$$G(s) = \frac{4.4 \times 10^{-7}}{(4s + 1)(30s + 1)}$$

Applying the Skogestad approximation converted this higher-order model to FOPTD form:

$$G_p(s) \approx \frac{4.4 \times 10^{-7} e^{-2s}}{32s + 1}$$

From this analytical model, the theoretical dynamic parameters are

$$K_{p,\mathrm{th}} = 4.4 \times 10^{-7}$$
$$\tau_{\mathrm{th}} = 32 \text{ s}$$
$$\theta_{\mathrm{th}} = 2 \text{ s}$$

The percent error between the simulation and analytical results was calculated using

$$\% \text{ error} = \left| \frac{\text{simulation} - \text{theoretical}}{\text{theoretical}} \right| \times 100$$

The process gain error can be calculated:

$$\% \text{ error in } K_p = \left| \frac{8.0 \times 10^{-7} - 4.4 \times 10^{-7}}{4.4 \times 10^{-7}} \right| \times 100$$
$$= \left(\frac{3.6 \times 10^{-7}}{4.4 \times 10^{-7}} \right) \times 100$$
$$\% \text{ error in } K_p = 81.8\%$$

The lag time error can be calculated:

$$\% \text{ error in } \tau = \left| \frac{5 - 32}{32} \right| \times 100$$
$$= \frac{27}{32} \times 100$$
$$\% \text{ error in } \tau = 84.4\%$$

The time delay error can be calculated:

$$\% \text{ error in } \theta = \left| \frac{0 - 2}{2} \right| \times 100$$
$$= 100\%$$

% error in $\theta = 100\%$

Parameter	From Excel Simulation	From Laplace/Skogestad Model	Percent Error
K_p	8.0×10^{-7}	4.4×10^{-7}	81.8%
τ	5 s	32 s	84.4%
θ	0 s	2 s	100%

The results show that the parameters extracted from the Excel dynamic simulation differ significantly from those obtained from the Laplace-transform-based FOPTD approximation. The primary reason for this discrepancy is that the analytical transfer function was derived from a linearized model and then further simplified using the Skogestad reduction. Both steps introduce approximation.

The time-domain simulation uses the original nonlinear ODE directly, so it preserves the full dependence of $k_L a$on airflow. In contrast, the transfer-function model assumes small deviations around the nominal operating point and neglects nonlinear interaction terms. This causes the theoretical gain and time constant to differ from the simulated values when the response is extracted from the nonlinear curve.

A second reason is the treatment of sensor dynamics. The analytical model explicitly includes a 30 s sensor lag, which strongly increases the apparent process time constant and introduces an effective delay after Skogestad reduction. If the Excel simulation is based only on the process ODE and does not explicitly model the sensor as an additional dynamic state, then the simulated response will naturally appear faster and show little to no dead time. This explains why the simulation produced $\tau \approx 5$s and $\theta \approx 0$s, while the analytical FOPTD model predicted $\tau = 32$s and $\theta = 2$s.

Finally, graphical extraction of $\tau$and $\theta$from a simulated response can introduce some reading uncertainty, especially when the response is fast and the time axis is coarse. Even so, the general trend is clear: the simulated system behaves much more quickly than the reduced FOPTD model predicts.

The Excel simulation confirms that the dissolved oxygen process responds in a stable first-order-like manner to a 10% increase in airflow. However, the dynamic parameters obtained from the simulation differ appreciably from the theoretical values derived from the Laplace-transform model with Skogestad approximation. The process gain from simulation was higher, the lag time was much shorter, and the time delay was essentially zero. These differences are attributable to nonlinear effects, omission or simplification of sensor lag in the simulation, and the approximate nature of the Skogestad reduction.

7 Increase on Inlet Air Flow

To evaluate the dynamic behavior of the dissolved oxygen system, a perturbation in the manipulated variable was introduced after the system reached steady state. A 10% step increase in the inlet air flow rate was applied at $t = 55$seconds. This perturbation enables the characterization of the transient response of the system and allows for the determination of key process parameters such as process gain, time constant, and time delay.

The nominal operating condition for the air flow rate is

$$F_{air,ss} = 500 \text{ ft3}/\text{min}.$$

A 10% increase corresponds to

$$\Delta F_{air} = 0.10 \times 500 = 50 \text{ ft3}/\text{min}.$$

Therefore, the air flow rate changes from 500 to 550 ft3/min at $t = 55$ seconds.

In deviation variable form, this input can be represented as

$$X(t) = 0 \text{ for } t < 55\text{seconds and } X(t) = 50 \text{ for } t \geq 55\text{seconds}$$

The mass transfer coefficient is directly dependent on the air flow rate and is given by the relation

$$k_L a = 0.25 + 0.001(F_{air} - 500)$$

Prior to the step change, the system operates at steady state with

$$k_L a = 0.25\text{s-1}$$

After the step increase in airflow, the mass transfer coefficient increases to

$$k_L a = 0.25 + 0.001(50) = 0.30\text{s-1}$$

This increase in $k_L a$eflects enhanced gas-liquid mass transfer due to increased bubble formation, improved mixing, and greater interfacial surface area within the reactor.

From the governing dynamic model it is evident that the mass transfer term $k_L a(C^* - C)$ represents the rate at which oxygen enters the liquid phase, while the term $K_{O_2}x$represents the rate of oxygen consumption by the cells. Since the biological consumption term remains constant during the perturbation, the increase in $k_L a$results in an increase in the net rate of oxygen accumulation within the reactor. Consequently, the dissolved oxygen concentration increases following the step change in airflow.

The dynamic simulation performed in Excel shows that prior to the step input, the system had reached steady state, as indicated by a constant dissolved oxygen concentration over time. At $t = 55$seconds, the step increase in airflow is introduced, and the dissolved oxygen concentration begins to increase immediately. The absence of any noticeable delay between the input change and the system response indicates that the time delay is approximately zero. This suggests that transport delays, actuator delays, and sensor lag are either negligible or not explicitly represented in the simulation model.

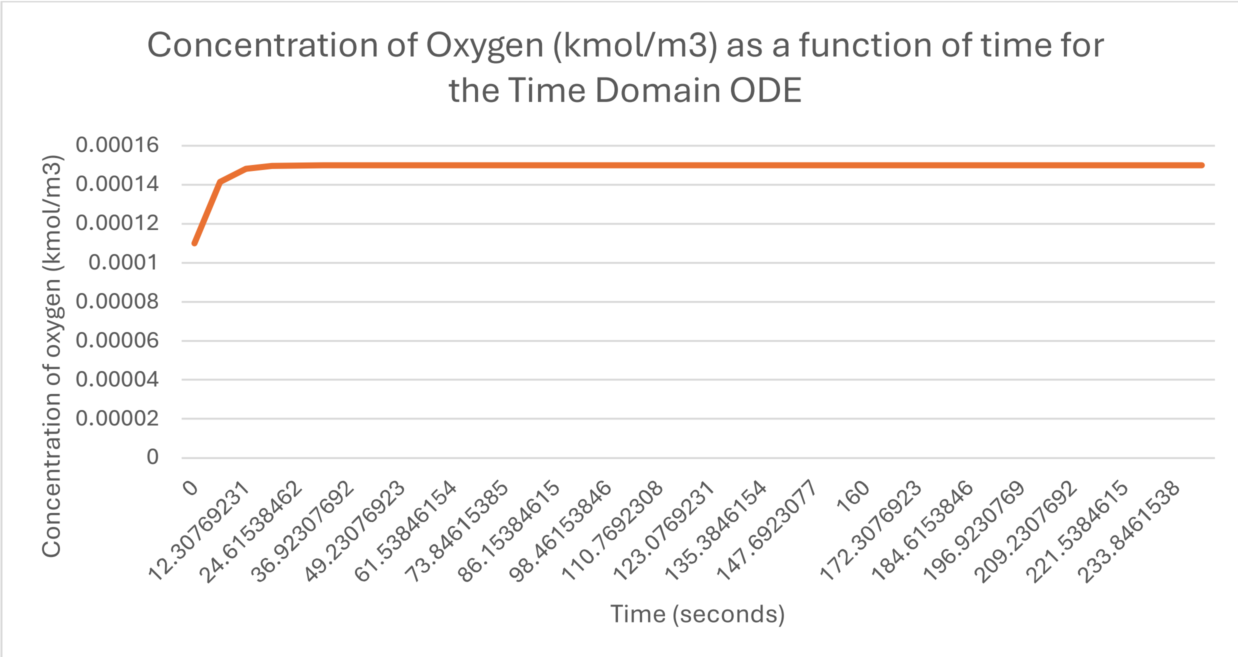
Following the step change, the dissolved oxygen concentration increases smoothly and asymptotically toward a new steady-state value. The shape of the response curve is characteristic of a first-order system, consistent with the linearized dynamic model derived earlier. The exponential nature of the response indicates that the system is dynamically stable and that the rate of approach to the new steady state is governed by a single dominant time constant.

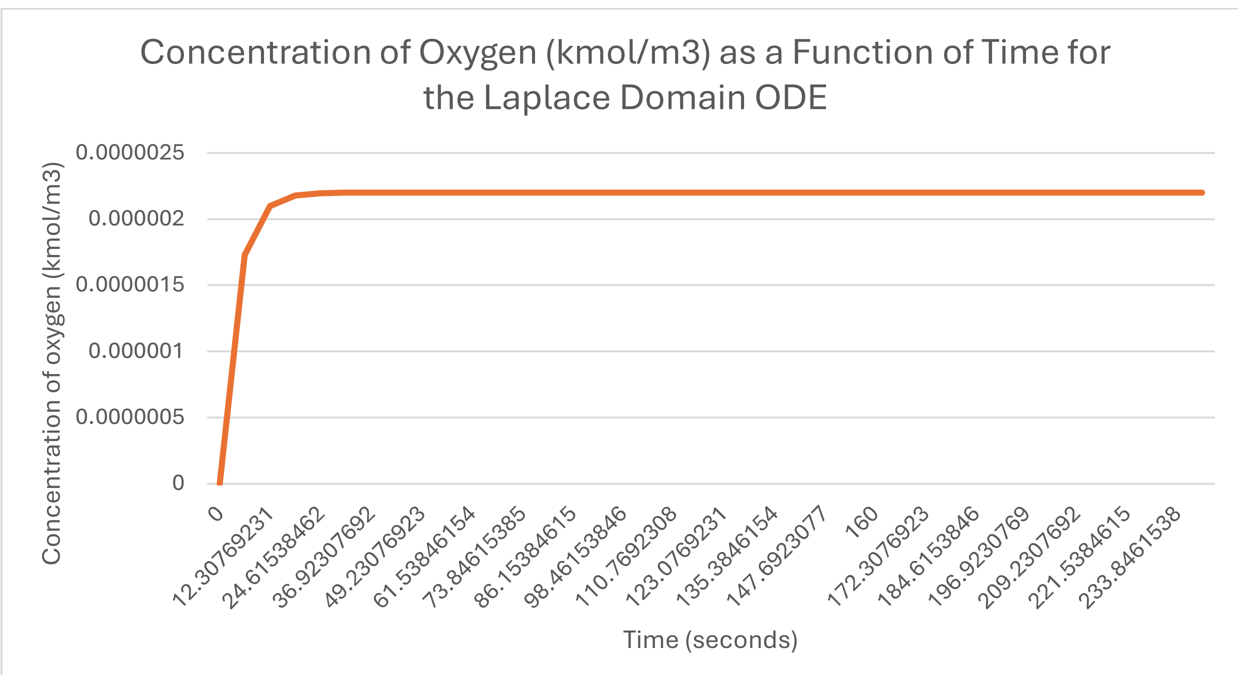
At the new steady state, the system once again satisfies the condition $k_L a(C^* - C) = K_{O_2} x$. However, because the mass transfer coefficient is now larger, the required driving force $(C^* - C)$is smaller. This results in a higher steady-state value of the dissolved oxygen concentration. Physically, this means that the system can maintain the same oxygen consumption rate with a smaller difference between saturation and actual concentration, due to enhanced transfer efficiency.

The observed response confirms that the dissolved oxygen concentration is positively correlated with the inlet air flow rate and that increasing airflow is an effective means of increasing oxygen availability in the liquid phase. The system exhibits fast, first-order dynamics with negligible delay, which differs from the behavior predicted by the Skogestad-reduced analytical model that includes an apparent time delay due to sensor dynamics. This discrepancy highlights the difference between the simplified transfer function approximation and the more detailed nonlinear simulation.

Overall, the step increase in inlet air flow provides a clear demonstration of the process dynamics and serves as the basis for extracting key parameters such as process gain, time constant, and time delay. The response obtained from this perturbation is essential for validating the analytical model and understanding the dynamic characteristics of the system.

8 Appendix





References:

Caracotsios, M. (n.d.). In *Transfer Functions in Dynamic Models*. essay.